Universal Design 2026: Advancing an Inclusive and Sustainable World for Everyone
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A study on universal design evaluation methods in an AI-powered environment: Based on experiments in Japan and Sweden focusing on user scanning behaviour

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Abstract. This study examines how optimising information presentation in personalised AI improves usability while also introducing confusion and anxiety due to differences in users' cognitive characteristics and cultural backgrounds.This study aims to re-examine methods for evaluating universal design (UD) common to diverse users by focusing on "scanning behaviour," which is how users search, select, and interpret information. To explore this issue, this study is based on a longitudinal analysis of UD evaluation reports by Japanese students and a purchasing behaviour experiment comparing students in Japan and Sweden. The results showed that, in addition to conventional elements such as visibility, the way users rely on visual cues changes depending on differences in information access conditions and experiences. These results indicate the need to view user diversity not as attributes, but as a process of information search and cognition, and provide a new perspective on universal design evaluation in the age of AI.

Keywords. Universal Design, Scanning behaviour, Purchasing Behaviour, Packaging Design, AI

1. Introduction

Universal Design (UD) is a concept that aims to create environments, products, and services that are easily accessible and user-friendly from the outset for as many people as possible, regardless of differences in age, ability, culture, language, etc. [1, 2]. With the adoption of the Convention on the Rights of Persons with Disabilities (CRPD) in

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2006, a global social trend of respecting human rights has progressed widely, and the importance of UD, which embraces diversity, has increased even further [3].

In recent years, artificial intelligence (AI) has contributed to the realisation of UD by providing information in diverse forms tailored to user preferences and situations. However, providing a large amount of detailed information may not always function appropriately for users with different cognitive characteristics, experiences, and cultural backgrounds, potentially causing confusion and anxiety.

Previous research has shown that even if objects are designed to be easy to see and understand, if there is an overload of information, users will recognise "cues" based on their own knowledge and behavioural habits [4–6]. In a future AI-driven society, it will be necessary to re-examine universal design evaluation indicators based on a classification of exploratory behaviour that transcends personal factors such as age, ability, and cultural/orientational characteristics.

Therefore, this study re-examines the evaluation method of UD by focusing on scanning behaviour in a series of actions related to product purchase decisions, based on hypotheses derived from previous research. In this study, scanning behaviour is defined as the process of visually and cognitively exploring, selecting, and interpreting multiple cues. Specifically, the aim is to verify the evaluation method of UD design using a hypothetical model of scanning behaviour and clarify future development directions through (1) an analysis of UD evaluation cases in shopping using conventional UD evaluation indicators, and (2) an analysis of qualitative research on exploratory behaviour regarding milk and confectionery packaging in Japan and Sweden.

2. Background

2.1 UD and Related Concepts

There are several design approaches other than UD that aim to "create a society that embraces diversity." For example, accessibility design adapts the environment and products based on guidelines and standards to meet the needs of individuals with specific functional characteristics [7]. Usability design, on the other hand, aims to achieve a "quality of experience" that is easy to use and efficiently achieves its purpose, through experiments involving stakeholders [8]. UD, on the other hand, aims to make things usable by more people without requiring post-design adaptation, and is guided by the seven principles advocated by Ron Mace. However, there are no specific methods or tools; it is realised through a broad, value-oriented approach that incorporates multiple design methods and tools [9].

2.2 Re-examining the Concept of UD considering Advances in AI Technology

Recent advancements in AI and other technologies are leading to a shift towards designs that allow users to select their preferred methods of use based on the situation and their preferences. This is also changing how we perceive human diversity. Instead of classifying people based on attributes such as age, ability, and cultural background, a spectral understanding [10] that views individual characteristics as a continuous and overlapping spectrum is becoming more widespread. For example, Panasonic's "Persona Spectrum Tool [11] " and Kajima Corporation's concept of "deeply understanding

people" demonstrate that human characteristics exist not as fixed categories but as a continuous spectrum.

This shift suggests that traditional design and evaluation frameworks based on differences in people's functional characteristics may not adequately capture users' information retrieval and decision-making processes in AI environments. Rather, what becomes important is the difference in how people interact with information, how they acquire, explore, and interpret it. Therefore, this research examines a UD evaluation method based on classification of exploration behaviour as a perspective beyond individual factors and verifies its effectiveness.

3. Conceptual Framework and Hypothesis

3.1 Overview Tool for Universal Design Evaluation

To facilitate the practical application and evaluation of Universal Design (UD), this study proposes a simplified visualisation tool that enables users to assess UD from two complementary perspectives:(1) the extent to which diverse users are considered, and (2) the range of UD design principles addressed.

The first graph assesses the inclusivity of human diversity across 10 user categories with differing physical abilities, cognitive abilities, and cultural/linguistic backgrounds (myself, children, the elderly, visually impaired, hearing impaired, lower body disabled, finger disabled, cognitive/memory impaired, mentally ill patients, and foreigners). Each category is rated on a five-point scale from “problematic” (1) to “very good” (5). The second chart evaluates the breadth of UD considerations based on nine criteria: fairness, flexibility, simplicity/clarity, intuitiveness, safety, physical effort, spatial usability, as well as economy and aesthetics. These are also assessed using a five-point scale. Together, these two radar charts enable a concise and intuitive overview of UD performance, making it easier for a wide range of users to conduct structured evaluations.

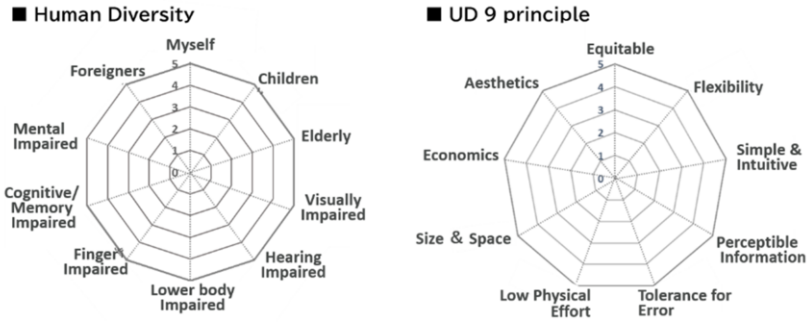


Figure 1. Two types of UD evaluation radar charts

3.2 Purchasing Behaviour Model

Based on previous research findings [5,12,13], this study visualises nine steps in the purchasing behaviour process and constructs a framework that allows users to understand and evaluate products across language and cultural differences. It is assumed that various design elements function as cues at each stage, and this study organises these into nine

items: (1) eye-tracking, (2) brightness, (3) colour, (4) visual information (symbols, illustrations, photographs, etc.), (5) textual information (numbers, product names, descriptions, etc.), (6) overall structure (size, shape, material, etc.), (7) user-friendliness, (8) preference and appeal, and (9) understanding how to use the product. These correspond to the step-by-step decision-making process (Figure 2).

Recent advances in AI technology have greatly improved support for initial recognition (image recognition, translation, speech synthesis, etc.). However, while these technologies enhance convenience and marketing effectiveness, they do not necessarily align with the principles of universal design, which emphasises autonomous information retrieval with minimal stress for all users.

Therefore, this study argues that in universal design evaluation, it is necessary to explicitly consider not only differences in physical and functional abilities, but also differences in the processes by which users acquire and search for information. Based on this perspective, we hypothesise that user diversity can be understood not only through conventional attribute-based classification, but also through variations in scanning behaviour.

This framework will serve as the basis for the scanning behaviour analysis in the following sections.

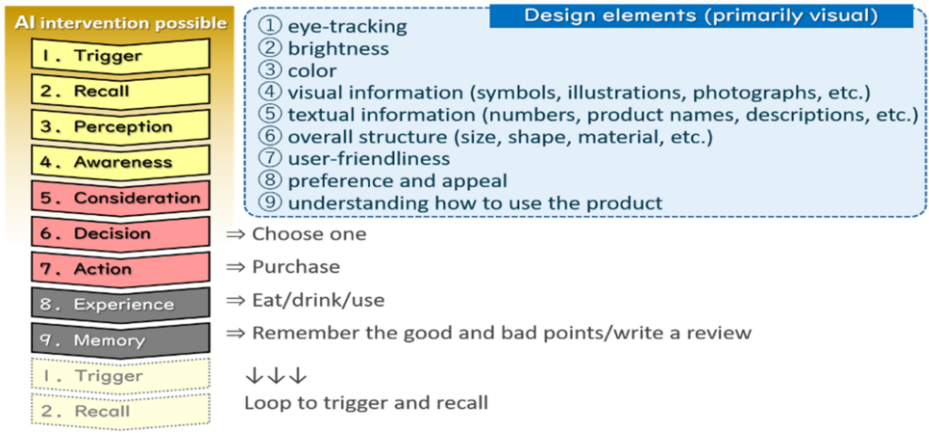


Figure 2. Purchase behaviour model

4. Methods

This study analyses how users obtain and process information during the purchasing decision-making process to test the hypothesis presented in Section 3. As a perspective for analysis, we focus on two modes of scanning behaviour: exploratory scanning (searching across multiple items) and focused scanning (scrutinizing specific items). These modes function as conceptual categories for interpreting the information acquisition process.

4.1 UD Evaluation Report Using Radar Charts in Shopping Behaviour

Students from design and cultural policy departments, around 20 years old, who participated in a UD class, researched "good examples" and "bad examples" in shopping

environments from a UD perspective. They evaluated these using a UD evaluation radar chart (Figure 1) and reported their observations with photographs and comments.

To verify the effectiveness of this evaluation method, a large dataset consisting of approximately 950 students and 3,000 samples collected between 2022 and 2025 was analysed. The analysis focused on identifying characteristic cases in the information acquisition process, interpreted from the perspectives of exploratory scanning and focused scanning.

4.2 Cross-Cultural Comparison of Scanning Behaviour in Purchase Decision-Making

A qualitative study was conducted to explore differences in information gathering in purchase decision-making across cultural backgrounds. Participants were presented with two types of Japanese snack foods and asked to list 1 to 5 pieces of information that influenced their purchase decision. They were given approximately 5 minutes to respond and instructed to explain what they focused on and what information they obtained.

This study was conducted in Sweden in September 2025 (n=29) and in Japan in November 2025 (n=37), targeting design students around 20 years old in both countries.

Furthermore, Swedish participants were presented with photographs of milk sections in Japan and Sweden, as well as examples of milk packaging. Participants were asked to list 1 to 5 pieces of information they considered when making a purchase decision.

In both studies, responses were collected using paper-based questionnaires and digital tools (e.g., Google Forms). The collected data was organised and analysed to identify information gathering patterns and cross-cultural differences and interpreted within the framework of scanning behaviour.

5. Results

5.1 Analysis of the "UD Evaluation Radar Chart" in Shopping Behaviour



Figure 3. Two examples of similar products receiving different evaluations

Using the UD Evaluation Radar Chart to compare highly-rated and low-rated items, it was confirmed that, in exploratory scanning, the ease of distinguishing between products in a series was an important factor. On the other hand, in focused scanning, the ease of

visually identifying illustrations and product contents, as well as the clear presentation of product volume and contents, were important factors.

These results indicate that the combination of multiple design elements influences the evaluation. Furthermore, it was confirmed that this evaluation tool allows for a quick understanding of items that are highly rated by many users and items that are perceived as issues. However, interpretation was difficult when the evaluation was intermediate, and such cases were frequently observed. In addition, it was confirmed that evaluations did not always match even for the same product (Figure 3).

5.2 Analysis of international differences in scanning behaviour

We organised the information students obtained, categorised by design elements, and analysed comments containing multiple elements.

As a result, Swedish students submitted 123 comments, while Japanese students submitted 235. The average number of comments per student was 4.2 and 6.4, respectively, with Japanese students submitting approximately 1.5 times more comments. Analysis by design element revealed that comments regarding the overall design, colour, and images were common to both groups. However, Japanese students also frequently mentioned detailed elements such as text, fonts, logos, and numbers (Figure 4).

Furthermore, because Swedish students did not understand Japanese, their access to information was limited, and they appeared to rely primarily on non-verbal and abstract information such as colour and overall impression. Consequently, they had less information, primarily engaged in exploratory scanning, and limited focused scanning. In contrast, Japanese students integrated both abstract and detailed information, employing both exploratory and focused scanning.

These results indicate that the type and accessibility of visually obtained information influence the amount of information acquired and processed during scanning behaviour.

The milk samples clearly demonstrated Sweden's culture of identifying contents by colour, a practice not found in Japan. Due to the wide variety of dairy products, detailed labelling of contents was required.

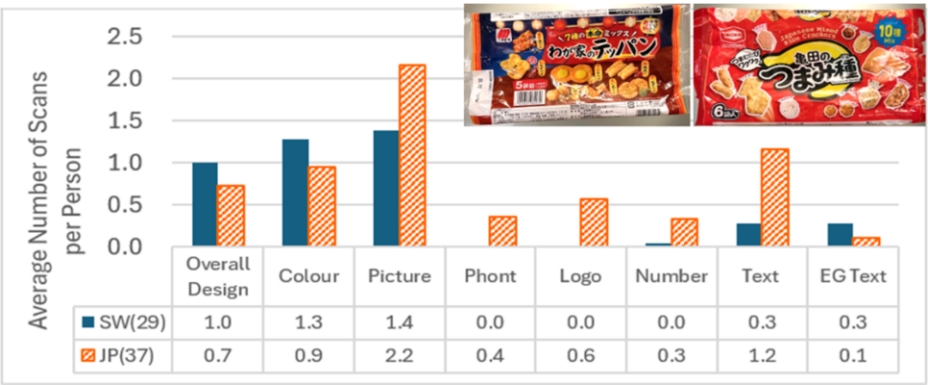


Figure4. Comparison of scanning behaviour for Japanese snack food packaging

6. Discussion

6.1 Variability and User Dependence in UD Evaluation

This study identified instances where evaluations of the same product were inconsistent. This result indicates that UD evaluation is not uniquely determined by fixed criteria, but rather depends on the user's experience, knowledge, and past usage history. Traditional UD evaluations relied on classifications based on attributes such as age and physical function, but the results of this study suggest that this framework alone may not adequately explain the differences in evaluations.

6.2 Positioning of Scanning behaviour

In this study, the initial stages of purchasing behaviour, from trigger, recall, perception, to cognition (awareness), were defined as scanning behaviours. This stage corresponds to preprocessing before the user recognises the object and functions as a prerequisite for subsequent consideration and decision-making. Therefore, the differences in behaviour observed in this study can be positioned primarily as differences in information processing up to this cognitive stage.

6.3 Differences in Information Processing in the Cognitive Stage

Swedish students, in situations where access to linguistic information was limited, relied on visual cues such as colour and images for cognition. On the other hand, Japanese students engaged in integrated information processing, including textual information and context. This difference can be interpreted not only as a difference in cultural background, but also as a difference in the conditions for accessing information.

6.4 Relationship with the Visual Search Characteristics of the Elderly

Previous research [5,6] has shown that the elderly tend to conduct sequential searches, checking individual elements one by one, rather than grasping an object from a holistic, overview perspective. Furthermore, when information is homogeneous, an increased cognitive load has been reported, such as the need to "follow each element with the eyes."

The behaviour of Swedish students observed in this study partially shares these search characteristics in that they rely on visual cues under circumstances where access to linguistic information is limited. This suggests that differences in scanning behaviour may arise not only from cultural background, but also from cognitive conditions such as limitations on access to visual and linguistic information.

6.5 Redefining UD Evaluation

Based on the above results, UD evaluation should be viewed not as a static classification based on user attributes, but as a process of information search and cognition. In other words, user differences should be described not by attributes such as "who they are," but by the process of "how they access and perceive information." This perspective offers important insights when considering flexible design approaches that can accommodate users with diverse cultural backgrounds and abilities.

Furthermore, by incorporating an understanding of how AI technology can be used to support and enable people’s different capabilities, one can gain critical insights when

considering flexible design approaches that accommodate users with diverse cultural backgrounds and abilities.

7. Conclusion

This study focused on UD evaluation in package design and scanning behaviour during purchase, analysing the user's information search and cognitive processes.

The results revealed that UD evaluation fluctuates depending not only on the physical characteristics of the product but also on the user's experience and cognitive background. Furthermore, it was confirmed that visual cues and information structure significantly influence the initial stages of cognition during scanning.

In addition, by comparing the condition of restricted access to linguistic information with the visual search characteristics observed in the elderly, it was suggested that differences in scanning behaviours can be interpreted not only as differences in cultural background but also as differences in cognitive conditions.

Moving forward, universal design needs to be redefined not only as a response to user attributes but also as a design that adapts to diverse scanning behaviour and cognitive conditions. We plan to investigate user scanning behaviour in relation to the adoption of AI technology, and to present a model that redefines design to suit these conditions.

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